

ROEBOURNE AP, Australia

WMO: 943080

Lat: 20.759S

Lon: 117.158E

Elev: 11

StdP: 101.19

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	11.0	-6.0	2.3	22.9	-4.1	2.7	22.7	11.4	22.1	10.5	22.5	1.9	210	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.8	42.5	22.0	41.1	22.1	39.9	22.4	27.2	33.8	26.8	33.1	26.5	32.7	6.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.2	29.8	25.4	20.6	29.6	24.9	20.0	29.5	86.5	34.0	84.5	33.2	82.7	32.6	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	10.1	9.3	6.9	45.9	1.6	1.5	5.7	47.0	4.8	47.9	3.9	48.8	2.7	49.9
			3.5	28.4	1.4	0.6	2.5	28.9	1.6	29.3	0.8	29.6	-0.3	30.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	31.9	32.0	31.3	28.9	24.1	20.6	20.1	21.2	24.1	28.3	30.3	31.7	(6)
(7)		DBStd	5.12	2.37	2.28	2.43	2.22	2.35	2.30	2.29	2.05	2.40	3.00	2.60	2.43	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	17	0	0	0	0	0	6	9	2	0	0	0	0	(9)
(10)		CDD10.0	6205	678	614	660	566	436	318	312	347	423	569	609	673	(10)
(11)		CDD18.3	3182	419	381	402	316	178	74	63	91	173	311	359	414	(11)
(12)		CDH23.3	42462	6019	5529	5780	4055	1779	607	662	1073	2057	4142	4879	5879	(12)
(13)		CDH26.7	24289	3629	3371	3466	2253	776	128	163	356	1022	2493	2989	3644	(13)
(14)	Wind	WSAvg	4.7	5.5	5.1	4.7	4.0	4.1	4.3	4.1	4.2	4.6	5.1	5.6	5.7	(14)
(15)	Precipitation	PrecAvg	296	58	78	50	16	26	35	14	6	1	0	1	9	(15)
(16)		PrecMax	656	327	301	323	114	147	309	85	27	8	5	13	100	(16)
(17)		PrecMin	52	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	154	86	78	72	28	36	62	22	9	2	1	2	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	43.8	42.8	40.1	35.8	31.2	31.9	33.5	37.9	41.9	43.2	44.1	(19)
(20)			MCWB	23.2	23.5	22.2	21.1	18.5	18.4	16.6	17.0	18.2	19.7	20.4	21.6	(20)
(21)		2%	DB	41.0	41.4	40.8	38.4	34.0	29.5	30.0	31.5	35.7	40.2	41.3	42.0	(21)
(22)			MCWB	24.3	23.9	22.4	20.8	18.5	16.9	16.3	16.7	17.6	19.7	20.7	21.6	(22)
(23)		5%	DB	39.2	39.5	39.1	36.9	32.5	28.2	28.4	30.1	34.0	38.5	39.6	40.1	(23)
(24)			MCWB	25.0	24.4	23.0	20.9	18.5	16.3	15.9	16.2	17.2	19.5	20.7	22.4	(24)
(25)		10%	DB	37.6	37.8	37.4	35.4	31.0	26.9	27.1	28.7	32.1	36.7	37.8	38.2	(25)
(26)			MCWB	25.1	24.4	23.2	20.9	18.2	16.2	15.5	15.7	17.1	19.6	20.6	22.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	28.1	27.4	26.5	24.5	23.6	21.7	20.7	21.5	24.2	25.7	27.1	(27)
(28)			MCDWB	36.2	34.7	32.9	31.1	27.0	25.4	24.8	27.0	28.7	31.9	32.6	34.8	(28)
(29)		2%	WB	27.2	27.3	26.8	25.5	23.1	21.8	19.9	19.5	20.4	23.0	24.8	26.3	(29)
(30)			MCDWB	34.2	33.7	32.0	30.3	27.5	24.5	24.8	26.5	28.1	31.6	31.8	33.0	(30)
(31)		5%	WB	26.7	26.9	26.3	24.8	21.8	20.3	18.6	18.4	19.6	22.1	23.9	25.6	(31)
(32)			MCDWB	33.4	33.1	31.6	29.9	26.8	24.3	24.4	25.9	27.8	31.1	31.3	32.4	(32)
(33)		10%	WB	26.4	26.4	25.9	24.0	20.7	18.9	17.3	17.4	18.8	21.3	23.1	25.0	(33)
(34)			MCDWB	32.9	32.7	31.3	30.0	26.2	23.7	24.0	25.4	27.6	30.9	31.8	32.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.3	10.8	10.9	12.9	13.4	12.8	14.3	15.8	16.0	15.6	14.0	12.4	(35)
(36)			MCDBR	13.2	13.8	14.0	14.8	15.4	14.2	16.1	17.6	18.2	17.7	16.3	15.0	(36)
(37)			MCWBR	3.9	4.0	4.5	5.0	5.8	5.8	6.5	6.9	6.5	5.8	5.1	4.6	(37)
(38)		5% WB	MCDWB	10.3	10.2	10.1	10.7	10.9	9.6	12.1	15.3	15.6	14.6	12.8	10.6	(38)
(39)	MCWBR		3.7	3.6	3.9	4.3	5.4	5.4	6.3	6.7	6.2	5.8	5.1	3.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.402	0.393	0.380	0.353	0.333	0.322	0.316	0.315	0.335	0.356	0.361	0.378	(40)	
(41)		taud	2.495	2.535	2.558	2.587	2.574	2.592	2.585	2.544	2.489	2.460	2.485	2.513	(41)	
(42)		Ebn at Noon	943	939	928	914	895	888	906	940	957	964	976	966	(42)	
(43)		Edh at Noon	116	110	103	93	87	82	85	95	109	117	117	114	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.44	7.11	6.36	5.56	4.66	4.21	4.58	5.65	6.68	7.55	8.04	8.00	(44)	
(45)		RadStd	0.46	0.49	0.46	0.29	0.35	0.30	0.29	0.16	0.25	0.18	0.21	0.34	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.46	N/A	N/A	N/A	+0.48	+0.56	N/A	N/A	N/A	+158

Nomenclature: See separate page